

Project 12: Flagship Capstone End-to-End Agentic Platform

Integrated Enterprise AI Workspace Powered by Google Gemini 2.5 Flash

1. Executive Overview

The Flagship Capstone Platform integrates Master Orchestrator Intent Planning, Specialist SQL Query Execution, Policy Safety Guardrails, and Synthesis Engines into a single unified workspace with complete observability trace logging.

2. System Architecture & Workflow Pipeline

- MasterPlannerAgent (Gemini 2.5 Intent Planning)
- SQLSpecialistAgent (Query Translation & Database Execution)
- PolicyGuardrailAgent (Safety Rules & Read-Only Checks)
- SynthesisEngine (Master Response Generation)
- Execution Trace Logger (Node Latency & Step Metadata)

3. API Endpoints Specification

Method	Endpoint	Description
GET	/	Serves Capstone Studio web UI
POST	/api/run-capstone	Executes master agentic workflow and returns trace log & synthesis

4. Deployment & Automated Verification

Standalone Uvicorn Launch:

```
cd 12_capstone/src && python -m uvicorn server:app --host 127.0.0.1 --port 8012
```

Automated Test Suite Execution:

```
python -m pytest 12_capstone/tests/
```